Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: P&D Dairy East

Physical address of dairy:

Number and Street	City	County	Zip Code
10860 S Ingomar Grade RD	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 10/01/1980

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X080-X041-XXXX X070-X080-X045-XXXX X070-X080-X075-XXXX X070-X080-X076-XXXX

B. OPERATORS

Lopes, Paul

Operator name: Lopes, Paul

Telephone no.: Landline: (209) 649-4967 Cellular:

Mailing Address Number and Street	City	State	Zip Code
9720 S Whitworth RD	Gustine	CA	95322

This operator is responsible for paying permit fees.

C. OWNERS

Lopes, Paul

Legal owner name: Lopes, Paul

Telephone no.: Landline: (209) 649-4967 Cellular:

Mailing Address Number and Street	City	State	Zip Code
9720 S Whitworth RD	Gustine	CA	95322

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	600	75	0	0	0	0
Maximum number	600	75	0	0	0	0
Average number	600	75	0	0	0	0
Avg live weight (lbs)	1,400	1,450	0	0		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 16,628.38 *tons per reporting period*

Total nitrogen from manure: 216,756.91 *lbs per reporting period*

After ammonia losses (30% loss applied): 151,729.84 *lbs per reporting period*

Total phosphorus from manure: 36,212.02 *lbs per reporting period*

Total potassium from manure: 115,693.94 *lbs per reporting period*

Total salt from manure: 299,756.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,640,000 *gallons*

Total nitrogen generated: 17,131.95 *lbs*

Total phosphorus generated: 2,932.20 *lbs*

Total potassium generated: 26,545.11 *lbs*

Total salt generated: 253,214.00 *lbs*

	5,640,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	5,640,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
F 26	60	60	1	process wastewater	X070-X080-X076-XXXX
F 28	21	21	1	process wastewater	X070-X080-X076-XXXX
F 29	21	21	1	process wastewater	X070-X080-X076-XXXX
F 30	17	17	0	none	X070-X080-X045-XXXX X070-X080-X046-XXXX
F 31	10	10	0	none	X070-X080-X046-XXXX
F 32	46	46	0	none	X070-X080-X041-XXXX X070-X080-X048-XXXX
Totals for areas that were used for application	102	102	3		
Totals for areas that were not used for application	73	73	0		
Land application area totals	175	175	3		

B. CROPS AND HARVESTS

F 26

Field name: F 26

05/01/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 60 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/04/2023	402.00 *ton*	As-is		59.4	9,420.00	2,190.00	18,900.00		17.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	6.70	126.23	29.35	253.26	952.07

F 28

Field name: F 28

F 28

06/15/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 21 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/10/2023	138.00 *ton*	As-is		59.4	10,400.00	2,080.00	21,600.00		21.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	6.57	136.69	27.34	283.89	1,152.58

F 29

Field name: F 29

06/15/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 21 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/10/2023	155.00 *ton*	As-is		56.4	12,000.00	1,680.00	18,100.00		24.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.38	177.14	24.80	267.19	1,570.43

NUTRIENT BUDGET

A. LAND APPLICATIONS

F 26 - 05/01/2023: Sudangrass, silage

Field name: F 26

Crop: Sudangrass, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	167.63	7,776,000.00 *gal*
Application event totals		0.00	0.00	0.00	167.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.68	9.36	84.72	808.13	1,080,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	149.01	6,912,000.00 *gal*
Application event totals		54.68	9.36	84.72	957.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.68	9.36	84.72	808.13	1,080,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	125.73	5,832,000.00 *gal*
Application event totals		54.68	9.36	84.72	933.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	158.32	7,344,000.00 *gal*
Application event totals		0.00	0.00	0.00	158.32	

F 26 - 05/01/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.60	8.32	75.31	718.34	960,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	121.07	5,616,000.00 *gal*
Application event totals		48.60	8.32	75.31	839.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	167.63	7,776,000.00 *gal*
Application event totals		0.00	0.00	0.00	167.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	139.70	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	149.01	6,912,000.00 *gal*
Application event totals		0.00	0.00	0.00	149.01	

F 28 - 06/15/2023: Sudangrass, silage

Field name: F 28

Crop: Sudangrass, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	186.26	3,024,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

F 28 - 06/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	159.65	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	159.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	172.96	2,808,000.00 *gal*
Application event totals		0.00	0.00	0.00	172.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.15	17.82	161.37	1,539.29	720,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	139.70	2,268,000.00 *gal*
Application event totals		104.15	17.82	161.37	1,678.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	172.96	2,808,000.00 *gal*
Application event totals		0.00	0.00	0.00	172.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	186.26	3,024,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	159.65	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	159.65	

F 28 - 06/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.43	11.88	107.58	1,026.20	480,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	133.04	2,160,000.00 *gal*
Application event totals		69.43	11.88	107.58	1,159.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	159.65	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	159.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	146.35	2,376,000.00 *gal*
Application event totals		0.00	0.00	0.00	146.35	

F 29 - 06/15/2023: Sudangrass, silage

Field name: F 29

Crop: Sudangrass, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	172.96	2,808,000.00 *gal*
Application event totals		0.00	0.00	0.00	172.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	186.26	3,024,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

F 29 - 06/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	159.65	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	159.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	146.35	2,376,000.00 *gal*
Application event totals		0.00	0.00	0.00	146.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.15	17.82	161.37	1,539.29	720,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	159.65	2,592,000.00 *gal*
Application event totals		104.15	17.82	161.37	1,698.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	133.04	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	133.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	106.43	1,728,000.00 *gal*
Application event totals		0.00	0.00	0.00	106.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	106.43	1,728,000.00 *gal*
Application event totals		0.00	0.00	0.00	106.43	

F 29 - 06/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.79	14.85	134.47	1,282.75	600,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	106.43	1,728,000.00 *gal*
Application event totals		86.79	14.85	134.47	1,389.18	

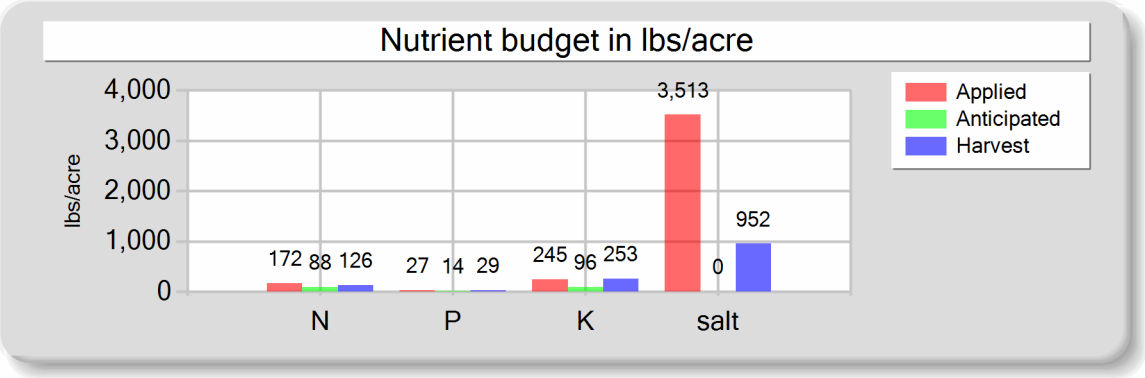
10/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	106.43	1,728,000.00 *gal*
Application event totals		0.00	0.00	0.00	106.43	

B. NUTRIENT BUDGET

F 26 - 05/01/2023: Sudangrass, silage

Field name: F 26 Crop: Sudangrass, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	157.95	27.03	244.74	2,334.60
Fresh water	0.00	0.00	0.00	1,178.10
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	171.95	27.03	244.74	3,512.69
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	126.23	29.35	253.26	952.07
Nutrient balance	45.73	-2.31	-8.52	2,560.62
Applied to removed ratio	1.36	0.92	0.97	3.69

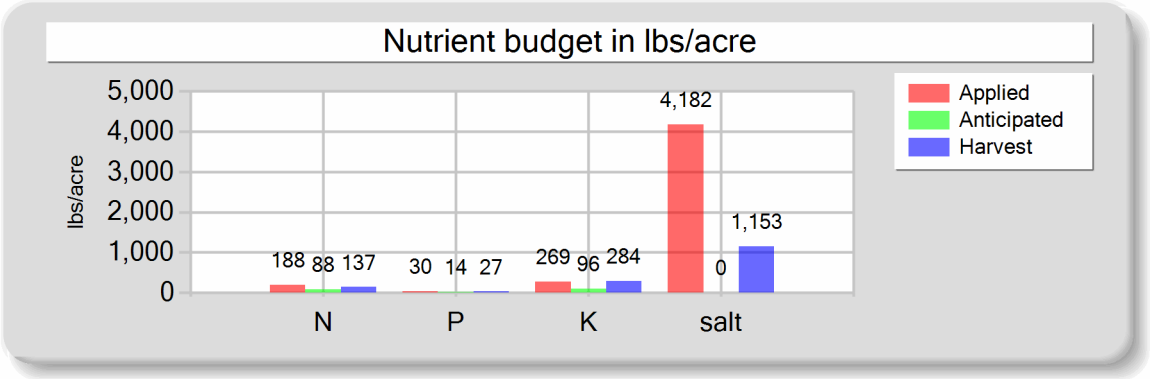
Fresh water applied	
54,648,000.00	*gallons*
2,012.50	*acre-inches*
33.54	*inches/acre*

Process wastewater applied	
3,120,000.00	*gallons*
114.90	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

F 28 - 06/15/2023: Sudangrass, silage

Field name: F 28 Crop: Sudangrass, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.58	29.71	268.95	2,565.49
Fresh water	0.00	0.00	0.00	1,616.47
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	187.58	29.71	268.95	4,181.97
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	136.69	27.34	283.89	1,152.58
Nutrient balance	50.89	2.37	-14.94	3,029.39
Applied to removed ratio	1.37	1.09	0.95	3.63

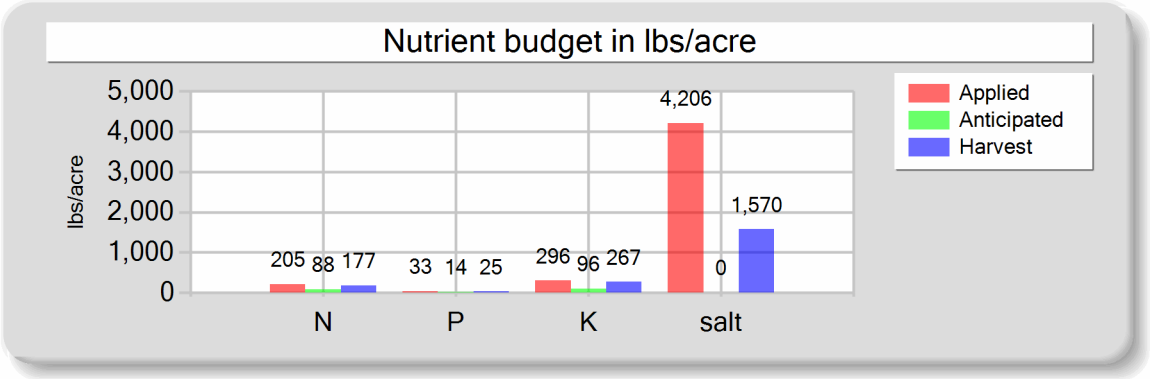
Fresh water applied	
26,244,000.00	*gallons*
966.48	*acre-inches*
46.02	*inches/acre*

Process wastewater applied	
1,200,000.00	*gallons*
44.19	*acre-inches*
2.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

F 29 - 06/15/2023: Sudangrass, silage

Field name: F 29 Crop: Sudangrass, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	190.93	32.68	295.84	2,822.04
Fresh water	0.00	0.00	0.00	1,383.65
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	204.93	32.68	295.84	4,205.69
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	177.14	24.80	267.19	1,570.43
Nutrient balance	27.79	7.88	28.65	2,635.26
Applied to removed ratio	1.16	1.32	1.11	2.68

Fresh water applied	
22,464,000.00	*gallons*
827.27	*acre-inches*
39.39	*inches/acre*

Process wastewater applied	
1,320,000.00	*gallons*
48.61	*acre-inches*
2.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall Solid Manure

Sample and source description: Fall Solid Manure

Sample date: 10/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,910.00	3,840.00	0.00	0.00	0.00	0.00	0.00		64.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ3

Sample and source description: WWQ3

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	364.00	210.00	0.00	0.00	62.30	564.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,830.00	5,380
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	885.00	510.00	0.00	0.00	90.70	1,180.00	244.00	219.00	468.00	4,280.00	0.00	243.00	0.00	9,310.00	8,680
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

CCID

CCID

CCID

Sample description: CCID

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

F 26 - 05/01/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 01/03/2024 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,420.00	2,190.00	18,900.00		17.50
DL	100.00	100.00	100.00		0.01

F 28 - 06/15/2023: Sudangrass, silage

F 28 - 06/15/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 01/03/2024 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	2,080.00	21,600.00		21.60
DL	100.00	100.00	100.00		0.01

F 29 - 06/15/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 01/03/2024 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	1,680.00	18,100.00		24.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

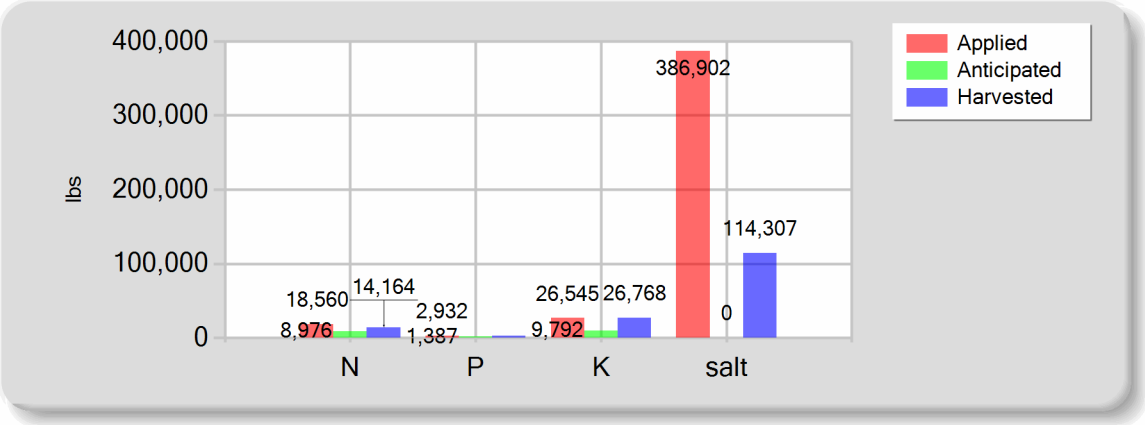
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

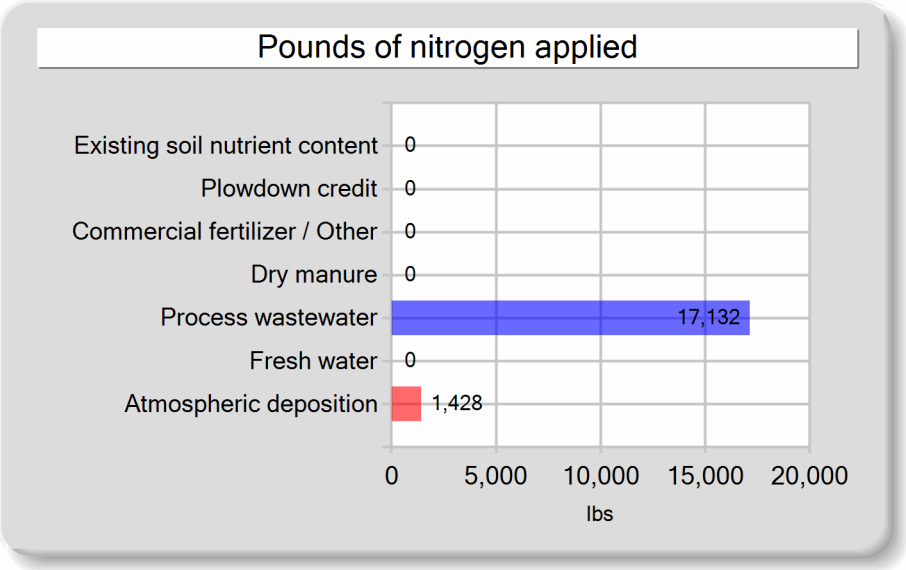
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

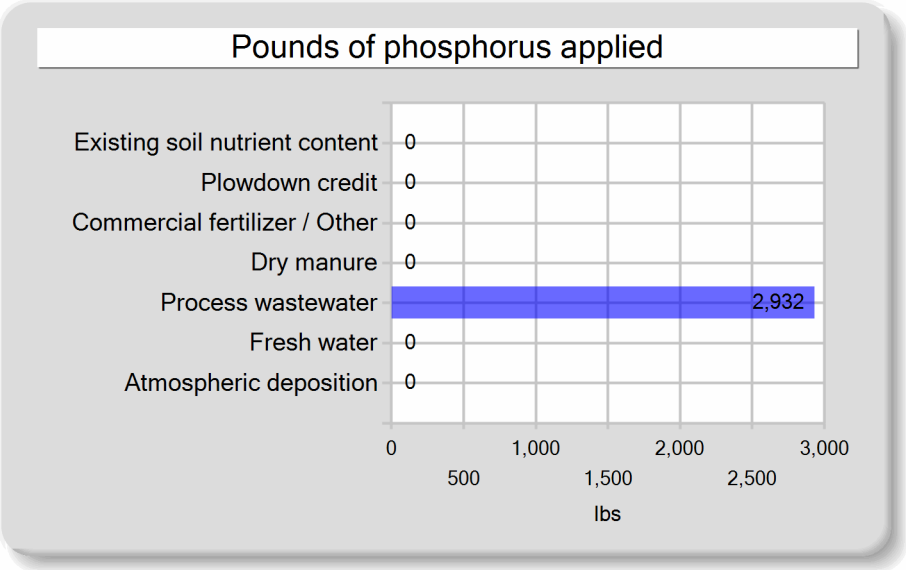
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	17,131.95	2,932.20	26,545.11	253,214.00
Fresh water	0.00	0.00	0.00	133,688.40
Atmospheric deposition	1,428.00	0.00	0.00	0.00
Total nutrients applied	18,559.95	2,932.20	26,545.11	386,902.41
Anticipated crop nutrient removal	8,976.00	1,387.20	9,792.00	0.00
Actual crop nutrient removal	14,164.08	2,855.64	26,768.20	114,307.34
Nutrient balance	4,395.87	76.56	-223.09	272,595.07
Applied to removed ratio	1.31	1.03	0.99	3.38

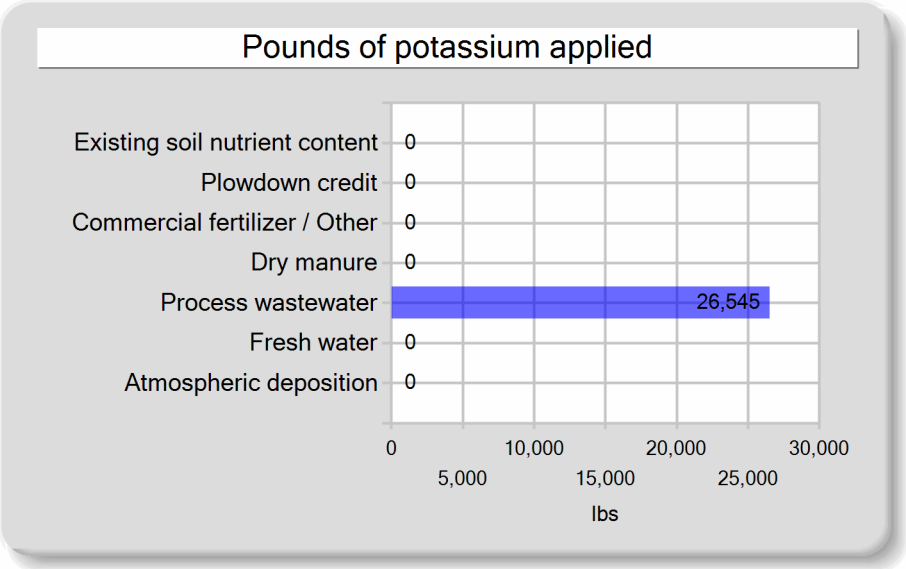
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

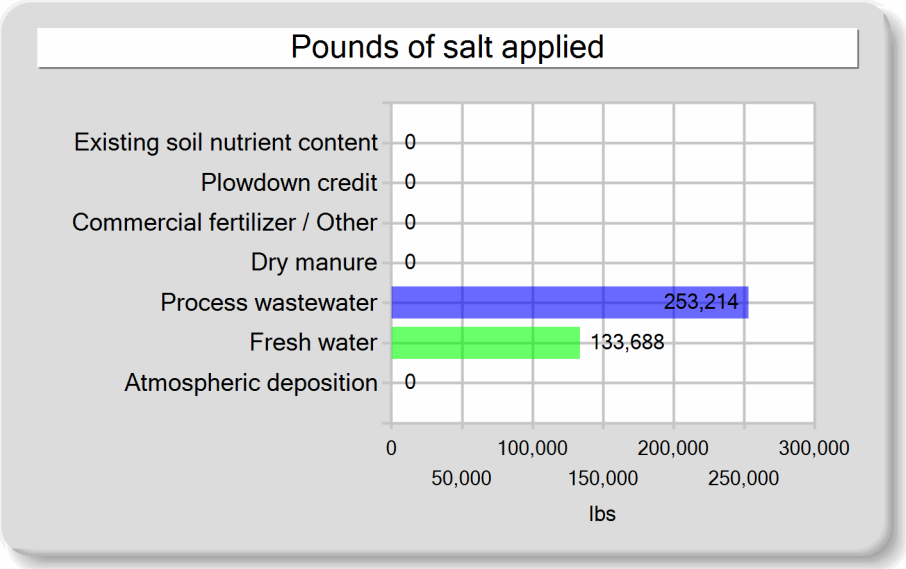


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Paul Lopes
Paul Lopes (Apr 18, 2024 23:47 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Paul Lopes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 18, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
P & D Dairy
9270 Whitworth Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: East
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW West Well	T4003006D	Electrical Conductivity (EC)	EPA 120.1 and	1.08	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.45	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	689	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)	0.619
Electrical Conductivity (EC)	99.2			1.05
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)	2.04

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
P & D Dairy
9270 Whitworth Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: East
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW East Well	T4003006E	Electrical Conductivity (EC)	EPA 120.1 and	1.23	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.987	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	787	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)	0.619
Electrical Conductivity (EC)	99.2			1.05
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)	2.04

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: P&D Dairy East
Mailing address: 9270 S. Whitworth Road
(not facility address) Gustine CA 95322
Phone#:

Authorized Copy Released To:
F&R

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	9/4		Field 26 Sudan							X	
2	11/10		Field 28 Sudan							X	
3	11/10		Field 29 Sudan							X	
4	1/2	3:30	West well	DW	12.0°C		X				
5	1/2	3:35	East well	DW	11.2°		X				
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
1st	Ja Ram					
2nd		DENELE ANALYTICAL, INC.	1-3-24 9:10			T4003006

(LABORATORY USE ONLY)

P&D East 2023 AR Complete

Final Audit Report 2024-04-19

Created:	2024-04-18
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAU9K9HSkXZ-WuqOmVzNbgGNbSIsI8nA__

"P&D East 2023 AR Complete" History

Document created by Joe Ramos (jramos@fragservices.com)
2024-04-18 - 6:21:25 PM GMT

Document emailed to pandd95@gmail.com for signature
2024-04-18 - 6:22:27 PM GMT

Email viewed by pandd95@gmail.com
2024-04-18 - 11:11:11 PM GMT

Signer pandd95@gmail.com entered name at signing as Paul Lopes
2024-04-19 - 6:47:56 AM GMT

Document e-signed by Paul Lopes (pandd95@gmail.com)
Signature Date: 2024-04-19 - 6:47:58 AM GMT - Time Source: server

Agreement completed.
2024-04-19 - 6:47:58 AM GMT